Anna Chen

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Education

University of California, Santa Cruz — **Starting September 2026**
Astrophysics PhD Candidate

Bowdoin College, Brunswick, ME — **May 2026**
Major: Physics **Minor**: Math — **GPA:** 3.94/4.00

Hanover High School, Hanover, PA — May 2022
Graduated Valedictorian — Unweighted GPA: 4.00/4.00

Research Experience

Computational Physics Summer Workshop — June 2026 - August 2026
Advisor: *Dr. Zachery Johnson*
Project: Average-atom Cassida Equation for Opacities in Matter at Extreme Conditions

Department of Physics, Bowdoin College, Brunswick, ME — December 2024 - May 2026
Advisor: *Dr. Thomas W. Baumgarte*
Project: Potential Dark Matter Trigger for Type 1a Supernova from Primordial Black Holes

- Operated Bowdoin HPC system to numerically simulate primordial black holes (PBH) through a white dwarf using BSSN formalism (C++)
- Wrote C++ class modeling white dwarfs with relativistic and nonrelativistic degenerate electron EOS to solve the Tolman-Oppenheimer-Volkoff Equations
- Wrote python script that determined how large a heated region must achieve to trigger a Type 1a Supernova
- Develop numerical code and writing senior thesis with autonomy that reinforce coding skills, leadership, and science communication

Department of Physics, Bowdoin College, Brunswick, ME — June 2024 - September 2024
Advsior: *Dr. Thomas W. Baumgarte*
Project: Constraining the nuclear equation of state from orbits of primordial black holes inside neutron stars

- Developed python class which modeled NSs using piecewise polytropic nuclear EOS to solve TOV equation
- Incorporated this model into code done by Baumgarte & Shapiro to compute precessing orbits of PBH within NSs
- Ran simulations of such orbits and observed that GW signals from these precessing orbits can further constrain the EoS for NSs
- Involved with writing manuscript published by Physics Review D

Harvard & Smithsonian Center for Astrophysics, Cambridge, MA — June 2023 - January 2024
Advisors: *Dr. Katharine Reeves, Dr. Xiaoyan Xie*
Project: Spatially Resolved Thermal Diagnostics of M1.0 Flare Observed by X-Ray Telescope (XRT) Aboard Hinode

- Conducted thermal diagnostics in python of a solar flare using the Filter-Ratio method on data taken by *Hinode*
- Gained insights into data analysis procedures and satellite data processes by analyzing spectroscopy data and making informed elemental abundance assumptions - notably, detection of coronal abundances helped identify the origin of ejected material.
- Contributed to the debugging of a newly developed Python package, XRTpy, resolving key oversaturation issues

Publications

- **Anna Chen**, Juan Diego DelPrado, Thomas W. Baumgarte, and Stuart L. Shapiro, 2024, "Constraining the nuclear equation of state from orbits of primordial black holes inside neutron stars," Phys. Rev. D 110, 064066 – Published 23 Sep. 2024 — Link to Preprint

- Xiaoyan Xie, **Anna Chen**, Crisel Suarez, Katharine K. Reeves, Soumya Roy, Nishu Karna, Christopher S. Moore, Joy Velasquez, "Disentangling Thermal and Abundance Characteristics in a Solar Flare Using SDO/AIA, Hinode/XRT, and MinXSS-1 Observations," *accepted to ApJ on 20, Oct. 2025* Link to ArXiv

Presentations and Posters

- **Bowdoin Presidential Poster Symposium**, Bowdoin College, Oct 2025 Link to Poster

  "Potential Dark Matter Triggers for Type 1a Supernova from Primordial Black Holes"

- **MIT Undergraduate Cosmology Workshop**, MIT, July 2025 Link to Presentation

  "Primordial Black Holes, White Dwarfs, and Type Ia Supernovae"

- **Eastern Gravity Meeting 2025**, University of New Hampshire, May 2025 Link to Presentation

  "Primordial Black Holes Inside Neutron Stars: Orbits, Precession, and Gravitational Wave Beats"

- **Bowdoin Presidential Poster Symposium**, Bowdoin College, Oct 2024 Link to Poster

  "Constraining the Nuclear Equation of State from Orbits of Primordial Black Holes Inside Neutron Stars"

- **MIT Undergraduate Cosmology Workshop**, MIT, July 2024 Link to Presentation

  "Constraining the Nuclear Equation of State from Orbits of Primordial Black Holes Inside Neutron Stars"

- **2024 AGU Fall Meeting Poster Session**, San Francisco, Dec 2024 Link to Poster

  "Thermal Diagnostics with Spatial Resolution of a Solar Flare Observed by the X-Ray Telescope (XRT) aboard *Hinode*"

- **CfA Solar REU Summer Symposium**, Center for Astrophysics, August 2023 Link to Presentation

  "Thermal Diagnostics with Spatial Resolution of a Solar Flare Observed by the X-Ray Telescope (XRT) aboard *Hinode*"

Awards

Burns Fellowship, Bowdoin College April 2025

- $6,000 plus $800 housing cost to conduct research over summer

Barry Goldwater Scholarship, National Scholarship March 2025

- $7,500 per year towards college tuition based on research merit

Book Award Winner, Bowdoin College Oct 2023

- Presented to every student who has earned a GPA of 4.00. Awarded freshman year.

Sarah and James Bowdoin Scholar, Bowdoin College Oct 2023

- Given to the top 20 percent of students, as calculated by GPA from the past year. Awarded freshman year.

Work Experience

Physics Learning Assistant, Bowdoin College January 2024 - Present

- Phys 1130: Introductory Physics I and Phys 1140: Introductory Physics II
- Taught mini lectures on topics relevant to lecture and lab during weekly learning group involving 5-15 students

Physics Grader, Bowdoin College January 2024 - Present

- Graded and provided feedback on student problem sets weekly which aided in exam preparation

Relevant Coursework

Physics:

Senior year: Honors Project in Numerical Relativity, Quantum Mechanics, Stars and Stellar Evolution, General Relativity (Spring 2025), Advanced Mechanics (Spring 2025), Statistical Physics (Spring 2025)
Junior year: Methods of Theoretical Physics, Methods of Computational Physics, Physics of Solids,

Electromagnetism, Physics of Black Holes (Audited)
Sophomore year: Introductory Astronomy, Electric Fields and Circuits, Engineering Physics, Quantum Physics and Relativity, Astrophysics: Galaxies and Cosmology
Freshman year: Introductory Physics I, Introductory Physics II

Mathematics:
Junior year: Introduction to Analysis
Sophomore year: Ordinary Differential Equations, Intro Mathematical Reasoning
Freshman year: Linear Algebra, Multivariate Calculus

Leadership and Co-curriculars

Women+ in Physics, Bowdoin College, *Leader* August 2024 - Present
- Organized scientist panels, volunteering opportunities for local schools, study nights, and merchandise
- Bridge communication between faculty and students by leading weekly meetings, sending emails, and propagating events
- Connected students through class years by facilitated a diverse and inclusive space in the physics community

Bowdoin Science Journal, Bowdoin College, *Managing Editor* January 2024 - Present
- Manages and edits articles for a small group of peers (4-5) in fields adjacent to my field of interest
- Co-founded a printed version of the Bowdoin Science Journal that now publishes in print once a semester
- Draw cover design that connects at the interface of science and art

Skills

Technical skills:
Proficient in Python, C++, Bash/Linux, Latex, Adobe InDesign
Familiar with Wolfram Mathematica and MATLAB
Language skills: Mandarin Chinese - Proficient